Zhonghan Wang

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Github: github.com/yogurt-shadow
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Education

Master student in Computer Science | ***Advised by: Bohua Zhan & Shaowei Cai*** — Sep. 2021 – Now
Institute of Software, Chinese Academy of Sciences

B.Eng. in Electronic Engineering (Rank top 15%) — Sep. 2017 – Jun. 2021
Nankai University

Publication

Efficient Local Search for Nonlinear Real Arithmetic (accepted) — VMCAI'24 (CCF-B)
Zhonghan Wang, Bohua Zhan, Bohan Li, Shaowei Cai. — 2024

Z3++ at SMT-COMP 2022 — SMT-COMP 2022
Shaowei Cai, Bohan Li, Jinkun Lin, **Zhonghan Wang** et al. (alphabetically ordered) — 2022

Work Experience

Research Intern — Oct 2022 – Aug 2023
Alibaba Group — Beijing, China

- Develop a new local search algorithm for end-to-end advertising allocation problem
- Our own local search solver is competitive with Gurobi and SCIP

Research Project

Solving SMT on nonlinear real arithmetic using MCSAT — Oct 2021
Institute of Software, Chinese Academy of Sciences (Supervised by Bohua Zhan & Shaowei Cai)

Using local search to solve high-order satisfiability problems — Jun 2022
Institute of Software, Chinese Academy of Sciences (Supervised by Bohua Zhan & Shaowei Cai)

Design local search algorithm for multilinear integer problem — Oct 2022
Alibaba Group

Honors and Awards

Distinguished Award of Tianjin College Mathematical Competition (Rank 17) — Summer 2018
Nankai University

Gold Winner of SMT-COMP 2022 (co-designer) — Fall 2022
Institute of Software, Chinese Academy of Sciences

Open Source Repository

Z3-plus-plus — https://github.com/shaowei-cai-group/z3pp
SMT Solver based on Z3 (Winner of SMT-COMP 2022 & 2023)

Z3-dnlsat — https://github.com/yogurt-shadow/z3-dnlsat
Dynamic MCSAT algorithm base on Z3

Skills

Languages: English (TOEFL 102, GRE 329 + 3.5), Chinese (Native)
Programming: Python (NumPy, SciPy, Matplotlib, Pandas), MATLAB, C & C++, Java
Document Creation: Microsoft Office Suite, Latex, Markdown